📋 Dokumentasi Lengkap Project nathBid

Platform Lelang Online Real-Time — Referensi konten untuk Video YouTube

📑 Daftar Isi

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1. 🎯 Gambaran Umum Project

nathBid adalah platform lelang online real-time yang dibangun dengan arsitektur modern full-stack. Aplikasi ini memungkinkan:

- **Membuat lelang** (sebagai penjual) dan **menawar barang** (sebagai pembeli)
- **Bidding secara real-time** menggunakan WebSocket — semua peserta melihat perubahan harga secara instan
- **Chat antar pengguna** secara real-time
- **Sistem keuangan lengkap** — top-up saldo, deposit bidding, invoice otomatis, reimbursement, pencairan dana
- **Admin panel** untuk approval lelang, verifikasi user, settlement, manajemen keuangan

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2. ⏳ Arsitektur & Tech Stack

> **Arsitektur:** Browser (React + TypeScript) ↔ Inertia.js ↔ Laravel 12 (PHP 8.2) ↔ PostgreSQL
> **Real-time:** Laravel Echo + Pusher.js ↔ Laravel Reverb WebSocket Server (port 8080)
> **Email:** Mailtrap SMTP │ **Build Tool:** Vite 7

Stack Backend (composer.json)

TEKNOLOGI	VERSI	FUNGSI
PHP	^8.2	Bahasa utama backend
Laravel	^12.0	Framework PHP utama
Laravel Reverb	^1.7	WebSocket server (first-party Laravel)
Laravel Fortify	^1.30	Authentication (login, register, 2FA, email verification)
Laravel Wayfinder	^0.1.9	Type-safe routing untuk frontend
Inertia.js (Server)	^2.0	Bridge Laravel ↔ React (SSR-capable)
Barryvdh DomPDF	^3.1	Generate PDF invoice server-side
PostgreSQL	—	Database utama

Stack Frontend (package.json)

TEKNOLOGI	VERSI	FUNGSI
React	^19.2.0	Library UI utama
TypeScript	^5.7.2	Static typing untuk JavaScript
Inertia.js (Client)	^2.3.7	SPA router — data dikirim sebagai props dari Laravel
Tailwind CSS	^4.0.0	Utility-first CSS framework
Vite	^7.0.4	Build tool & dev server
Laravel Echo	^2.3.0	Client-side library untuk menerima broadcast WebSocket
Pusher.js	^8.4.0	Protocol transport untuk Echo (kompatibel Reverb)
Radix UI	Various	Headless UI primitives (dialog, dropdown, avatar, dll.)
Lucide React	^0.475.0	Icon library
Sonner	^2.0.7	Toast notification library
class-variance-authority	^0.7.1	Variant-based styling utility
clsx + tailwind-merge	—	Conditional CSS classes
html2pdf.js	^0.14.0	Client-side PDF generation (fallback)
input-otp	^1.4.2	OTP input component
Headless UI	^2.2.0	Un-styled accessible UI components
tw-animate-css	^1.4.0	Animation utilities untuk Tailwind
React Compiler (babel)	^1.0.0	Auto-memoization dari React Compiler

Dev Tools

TOOL	FUNGSI
ESLint	Linting JavaScript/TypeScript
Prettier	Code formatting
Laravel Pint	PHP code style fixer
Laravel Pail	Real-time log viewer
PHPUnit	Unit testing
Concurrently	Menjalankan multiple processes sekaligus

3. 📡 Laravel Reverb — WebSocket Real-Time

> **Laravel Reverb** adalah WebSocket server bawaan (first-party) dari Laravel. Berbeda dengan Pusher (cloud-based), Reverb berjalan di server sendiri:
> ✅ **Gratis** — tidak perlu bayar third-party | ✅ **Self-hosted** — data di server sendiri | ✅ **Low latency**

Konfigurasi (.env)

```
BROADCAST_CONNECTION=reverb
REVERB_HOST="localhost"
REVERB_PORT=8080
REVERB_SCHEME=http
```

Frontend — Koneksi WebSocket (resources/js/echo.ts)

```
import Echo from 'laravel-echo';
import Pusher from 'pusher-js';

window.Pusher = Pusher;  // Pusher.js sebagai protocol layer
window.Echo = new Echo({
    broadcaster: 'reverb',  // koneksi ke Reverb, bukan Pusher cloud
    key: import.meta.env.VITE_REVERB_APP_KEY,
    wsHost: import.meta.env.VITE_REVERB_HOST,
    wsPort: import.meta.env.VITE_REVERB_PORT ?? 80,
    forceTLS: false,
    enabledTransports: ['ws', 'wss'],
    authEndpoint: '/broadcasting/auth',
});
```

> **Catatan:** Meskipun import `pusher-js`, koneksi diarahkan ke **Reverb** (self-hosted). Pusher.js hanya digunakan sebagai protocol transport karena Reverb kompatibel dengan Pusher protocol.

3 Broadcast Channels (routes/channels.php)

CHANNEL	TIPE	FUNGSI
`App.Models.User.{id}`	Private	Notifikasi pribadi per user
`chat.{id}`	Private	Chat real-time antar 2 user
`auction.{id}`	Presence	Lelang real-time — peserta bisa saling lihat siapa online

5 Event Broadcasting

EVENT	CHANNEL	TRIGGER	DATA YANG DIKIRIM
BidPlaced	auction.{id} Presence	Bid baru masuk	Harga baru, nama bidder (anonim), total bids, sisa waktu
AuctionTimeExtended	auction.{id} Presence	Bid di 10 detik terakhir	Waktu baru ends_at, +15 detik
AuctionEnded	auction.{id} Presence	Lelang berakhir	Harga final, nama pemenang, status ended
ActivityPosted	auction.{id} Presence	Chat/log di "War Room"	Tipe activity, konten, nama user
MessageSent	chat.{id} Private	Pesan chat dikirim	Konten pesan, info sender

> Semua event mengimplementasi `ShouldBroadcastNow` — broadcast langsung tanpa masuk queue.

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4. ⏰ Laravel Artisan Schedule

File: `routes/console.php` — Dijalankan dengan: `php artisan schedule:work`

SCHEDULE	FREKUENSI	FUNGSI
`auctions:update-statuses`	Setiap menit	Cek lelang yang harus mulai (scheduled → live) dan berakhir (live → ended, dispatch ProcessAuctionEnd)
`CheckPaymentDeadline` (Job)	Setiap menit	Cek invoice expired (lewat 1 jam), pindahkan ke runner-up atau batalkan
`ReimbursementService::markEligible()`	Setiap jam	Tandai reimbursement yang lewat masa tunggu 3 hari → eligible
`AuctionSettlementService::processScheduledDisbursements()`	Setiap 5 menit	Cairkan dana settlement yang dijadwalkan ke saldo seller

Dev Script (composer dev)

```
composer dev  →  Menjalankan 4 proses bersamaan:
1. php artisan serve         — HTTP server
2. php artisan queue:listen  — Queue worker
3. php artisan pail          — Real-time log viewer
4. npm run dev               — Vite dev server

# Terpisah di terminal lain:
php artisan reverb:start     — WebSocket server
php artisan schedule:work    — Scheduler
```

5. 🎯 Alur Aplikasi End-to-End

A. Alur Registrasi & Autentikasi

1️⃣ User isi form registrasi → POST /register (via Laravel Fortify)

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2️⃣ Backend simpan user + generate PIN unik 6 digit → Kirim email OTP via Mailtrap

↓

3️⃣ User masukkan kode OTP → POST /otp/verify → email_verified_at diisi

↓

4️⃣ Redirect ke Dashboard ✅

Fitur Auth: Login, Register, Reset Password, Email Verification (OTP), 2FA (Authenticator App), Session di database

B. Alur Pembuatan Lelang (Seller)

1️⃣ Seller buat lelang (judul, deskripsi, gambar, harga awal, bid increment, dll)

↓

2️⃣ Disimpan dengan status `pending` → Muncul di Admin Panel

↓

3️⃣ Admin approve → status `scheduled` + set starts_at & ends_at

↓

4️⃣ Scheduler deteksi starts_at ≤ now() → status `live` → Muncul di daftar lelang

Status Lifecycle: `pending → scheduled → live → ended / cancelled`

C. Alur Bidding Real-Time (Inti Aplikasi) ⭐

1️⃣ User buka halaman lelang → join **Presence Channel** auction.{id}

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2️⃣ Klik Quick Bid (+1M) → POST /auctions/{id}/bid → **Snap Lock 1 detik** (anti-spam)

↓

3️⃣ `DB::transaction + lockForUpdate()` → Validasi (is live? amount valid? bukan seller? saldo cukup?)

↓

4️⃣ **Deposit-Hold Model:** Hanya charge SELISIH (bid baru - deposit lama) dari saldo user

↓

5️⃣ Update current_price, total_bids → Cek apakah di **10 detik terakhir** → extend **+15 detik**

↓

6️⃣ `broadcast(BidPlaced)→toOthers()` → Semua peserta lain melihat harga berubah secara real-time

6 Mekanisme Kunci Bidding:

1. **Snap Lock (1 detik)** — `Cache::add()` mencegah spam bid
2. **DB Transaction + lockForUpdate()** — Mencegah race condition
3. **Deposit-Hold Model** — User hanya dicharge selisih bid baru vs deposit lama
4. **Increment System** — Bid harus kelipatan dari `bid_increment`
5. **Extended Time** — Bid di 10 detik terakhir → timer +15 detik (seperti lelang profesional)
6. **Quick Bid Buttons** — Tombol cepat (+1x, +2x, +5x, +10x dari increment)

D. Alur Pasca-Lelang (Timer = 00:00)

1️⃣ Scheduler `auctions:update-statuses` → Dispatch `ProcessAuctionEnd` Job

↓

2️⃣ DB transaction: Tentukan pemenang (bid tertinggi) → Buat Invoice (deadline 1 jam)

↓

3️⃣ Generate Reimbursement untuk peserta kalah → **Auto-approve** → **Kembalikan saldo**

↓

4️⃣ Buat AuctionSettlement (untuk pencairan ke seller)

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5️⃣ `broadcast(AuctionEnded)` → Notifikasi: ⏳ Winner | 💸 Refund losers | 💰 Seller sold

E. Alur Pembayaran Winner

Winner buka halaman Invoice → Countdown 1 jam → Bayar dari saldo → Invoice **paid**

⚠️ Jika 1 jam tanpa bayar:

`CheckPaymentDeadline` → Invoice **expired** → Cari runner-up → Buat invoice baru (atau cancel)

F. Alur Settlement & Pencairan Dana Seller

Settlement dibuat otomatis → Admin approve → Seller konfirmasi kirim → Admin mark delivered

↓

Scheduler cairkan dana ke saldo seller (amount - app fee Rp 20.000)

↓

Seller bisa tarik ke bank via Disbursement (fee 2.5%)

6. 🗄️ Database: Skema & Relasi Antar Tabel

Database: PostgreSQL (port 5432) | **Total Tabel:** ~21 tabel

Tabel Utama & Relasi

TABEL	KOLOM PENTING	RELASI
users	name, email, pin (6-digit), balance, role_id, verification_status	→ roles, → auctions(seller/winner), → bids, → invoices, → topups, → disbursements, → bank_accounts, → transactions, → chats
roles	name (superadmin/admin/user), display_name	← users
auctions	uuid, title, images(JSON), starting_price, current_price, bid_increment, registration_fee, starts_at, ends_at, original_ends_at, status	→ users(seller, winner), → auction_categories, ← bids, ← invoices, ← participants, ← deposits, ← settlements, ← activities, ← views
bids	auction_id, user_id, amount, is_winning, is_auto_bid, ip_address	→ auctions, → users
invoices	uuid, invoice_number, amount, admin_fee, registration_fee, app_fee, total_amount, status, payment_due_at	→ auctions, → users(winner)
auction_participants	auction_id, user_id, fee_paid, status (active/refunded)	→ auctions, → users
auction_deposits	auction_id, user_id, amount (dana ditahan)	→ auctions, → users
auction_settlements	amount, admin_fee(0%), app_fee(Rp20k), seller_amount, status, fund_status, delivery_status	→ auctions, → users(seller, winner, approved_by)
reimbursements	uuid, amount, status, eligible_at, reference_number	→ auctions, → users, → auction_participants
topups	amount, payment_method, status, virtual_account_number, expired_at	→ users
disbursements	amount, fee(2.5%), total, status, reference_number	→ users, → bank_accounts
bank_accounts	bank_code, account_number, account_name, is_primary	→ users, ← disbursements
transactions	type, amount, description, reference_id/type (polymorphic)	→ users, → any model (polymorphic)
chats	user_one_id, user_two_id	→ users(x2), ← messages
messages	chat_id, sender_id, content, is_read	→ chats, → users
auction_activities	type (bid/chat/system), content, metadata(JSON)	→ auctions, → users
auction_categories	name, slug, description, icon	← auctions
auction_durations	minutes, label	—
email_otps	email, otp, expires_at	—
notifications	type, notifiable_type/id, data(JSON), read_at	→ users (polymorphic)

7. 👥 Sistem Role & Hak Akses

ROLE	NAMA	HAK AKSES
superadmin	Super Admin	Akses penuh: verifikasi user, kelola admin, semua fitur admin
admin	Admin	Approval lelang, settlement, reimbursement, disbursement
user	User	Buat lelang, bidding, top-up, tarik dana, chat

Middleware: `CheckRole` (cek role), `HandleInertiaRequests` (shared data), `UpdateUserLastSeenAt` (online status)

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8. 📱 Halaman Frontend (React Pages)

Halaman User

- **Dashboard** — Ringkasan saldo
- **Daftar Lelang** — Browse (Live, Coming Soon, Selesai)
- **Detail Lelang** — ⭐ Bidding real-time + War Room
- **My Auctions** — Lelang yang dibuat (seller)
- **Chat** — Real-time chat antar user & admin
- **Top Up** — BCA VA, OVO, GoPay, DANA, QRIS
- **Disbursement** — Tarik saldo ke bank
- **Bank Accounts** — Kelola rekening
- **Transactions** — Riwayat transaksi
- **Invoices** — Lihat & bayar + download PDF
- **Settings** — Profil, password, 2FA

Halaman Admin

- **Auction Approval** — Review & approve/reject
- **User Verification** — Approve/reject user baru
- **Categories** — Kelola kategori lelang
- **Durations** — Kelola preset durasi
- **Settlements** — Approve + tracking pengiriman
- **Disbursements** — Approve penarikan saldo
- **Reimbursements** — Approve pengembalian dana

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9. 💰 Sistem Keuangan

Top Up → Masuk **User Balance**

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Bayar **Biaya Registrasi** + **Bid Deposit** (hold saldo)

↓ Lelang berakhir

Kalah: Auto Reimbursement → kembali ke saldo | **Menang:** Invoice dibuat (deadline 1 jam)

↓ Bayar invoice

Dana ditahan di **Settlement** → Admin approve + barang delivered → **Pencairan ke Seller**

↓

Seller: **amount - app_fee (Rp 20.000)** → Bisa tarik via **Disbursement** (fee 2.5%)

Tipe Transaksi

💚 Income (Masuk)

- `topup` — Isi saldo
- `bid_refund` — Return deposit bid
- `reimbursement` — Return biaya registrasi
- `settlement_seller` — Pencairan penjualan
- `settlement_refund` — Return settlement
- `disbursement_refund` — Return disbursement gagal

🔴 Expense (Keluar)

- `registration_fee` — Biaya registrasi
- `bid` — Deposit bid
- `disbursement` — Penarikan saldo

10. 🔧 Service Layer & Notifikasi

5 Service Classes

SERVICE	FUNGSI
BiddingService	Logika bidding: validasi, snap lock, deposit-hold, time extension, broadcast
TopupService	Generate virtual account / QRIS, simulasi pembayaran, update saldo
DisbursementService	Validasi penarikan, hitung fee 2.5%, proses pencairan
ReimbursementService	Generate reimbursement untuk peserta kalah, mark eligible, auto-approve
AuctionSettlementService	Buat settlement, approve, jadwalkan + proses pencairan dana ke seller

4 Notification Classes (Database Channel)

NOTIFICATION	TRIGGER
AuctionWonNotification ⏳	Pemenang lelang ditentukan
AuctionLostNotification 💸	Peserta kalah + reimbursement otomatis
AuctionSoldNotification 💰	Seller: barangmu terjual
InvoiceExpiredNotification ⚠️	Invoice lewat deadline 1 jam

Email: Mailtrap SMTP (sandbox) — `OtpMail` class untuk kirim OTP verifikasi email

PDF: `barryvdh/laravel-dompdf` (server-side) — Generate invoice PDF yang bisa didownload

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11. 🚀 Cara Menjalankan Project

```
# Setup Awal
composer install && npm install
cp .env.example .env
php artisan key:generate
php artisan migrate
php artisan db:seed

# Terminal 1: Laravel + Queue + Vite
composer dev

# Terminal 2: WebSocket Server
php artisan reverb:start

# Terminal 3: Scheduler
php artisan schedule:work
```

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12. 📊 Ringkasan Angka-Angka Penting

METRIK	JUMLAH
Model Eloquent	**21**
Controller	**24** (16 user + 8 admin)
Service	**5**
Event (Broadcast)	**5**
Job (Queue)	**2**
Notification	**4**
Artisan Command	**2**

METRIK	JUMLAH
Scheduled Task	**4**
Migration	**36**
Seeder	**6**
React Pages	**~25+**
Broadcast Channel	**3**
Tabel Database	**~21**
Middleware Custom	**5**

🏷️ Tagline untuk Video YouTube

"nathBid — Membangun Platform Lelang Online Real-Time dengan Laravel 12, React 19, TypeScript, Laravel Reverb WebSocket, dan PostgreSQL. Full-stack modern web application dengan fitur bidding real-time, sistem keuangan lengkap, dan admin panel."

Keywords SEO: Laravel 12 · React 19 · TypeScript · Inertia.js · Laravel Reverb · WebSocket · Real-time Bidding · PostgreSQL · Tailwind CSS v4 · Full Stack · Online Auction · Laravel Echo · Presence Channel · Race Condition · Database Transaction · lockForUpdate · Artisan Schedule · Queue Jobs